

Lagos Flood Risk Monitoring Dashboard

Detailed ArcGIS Pro procedure — Steps 4 through 7 (Drainage & Infrastructure, Flood Risk Index, Population Overlay, Hotspot Analysis).

Assumptions carried forward from Steps 1–3

- bullet Project Lagos_Flood_Risk is already created in ArcGIS Pro with a file geodatabase Lagos_Flood_Risk.gdb.
- bullet Lagos state and LGA administrative boundaries are loaded (feature class Lagos_Boundary) in **WGS 1984 UTM Zone 31N** (EPSG:32631). All layers should share this projected CRS for metric measurements (buffers, zonal stats).
- bullet A filled, hydrologically corrected DEM (Lagos_DEM_Fill) and derived slope / flow-accumulation rasters exist from Step 2.
- bullet Reclassified rainfall raster (Rainfall_reclassified, values 1 / 2 / 3) exists from Step 3, clipped to the Lagos boundary.
- bullet The **Spatial Analyst** extension is enabled (*Project > Licensing > Configure Licensing > Esri Extensions*).

Cell size / snap raster convention: set a single environment for the entire project so every raster output aligns pixel-for-pixel. This matters a lot in Step 5.

Go to *Analysis > Environments* and set:

- bullet **Current & Scratch Workspace** → Lagos_Flood_Risk.gdb
- bullet **Output Coordinate System** → WGS 1984 UTM Zone 31N
- bullet **Processing Extent** → Lagos_Boundary
- bullet **Cell Size** → 30 m (match SRTM)
- bullet **Snap Raster** → Lagos_DEM_Fill
- bullet **Mask** → Lagos_Boundary

Step 4 — Drainage & Infrastructure Data

Goal: bring Lagos roads, rivers, waterways and drainage lines into the project, isolate drainage from roads, and create 50 m / 100 m proximity (buffer) zones that will feed the flood risk index.

4.1 Download OSM data from HDX / QuickOSM

- 1 Open a browser and go to **data.humdata.org**. Search *Nigeria OpenStreetMap roads* and *Nigeria OpenStreetMap waterways*. Download the latest shapefile ZIPs (hotosm_nga_roads_lines_shp.zip and hotosm_nga_waterways_lines_shp.zip).
- 2 Unzip into a folder, e.g. C:\GIS\Lagos\OSM\.
- 3 **Alternative (recommended inside ArcGIS Pro):** install the *QuickOSM* add-in from ArcGIS Living Atlas, or use the *OSMnx* Python package. With QuickOSM you can query only the Lagos bounding box, which is much smaller than the whole country.

Typical QuickOSM / Overpass query (keys you want):

```
highway=*      waterway=river|stream|drain|canal|ditch    natural=water
landuse=reservoir
```

4.2 Load and re-project the OSM layers

- 1 In the **Catalog** pane, drag the roads and waterways shapefiles into the current map. They arrive in WGS84 geographic (EPSG:4326).
- 2 Run **Project** (Data Management Tools > Projections and Transformations) on each layer. Output CRS = *WGS 1984 UTM Zone 31N*. Save outputs into *Lagos_Flood_Risk.gdb* as *OSM_Roads_UTM* and *OSM_Waterways_UTM*.
- 3 Clip both layers to Lagos: **Analysis > Clip** using *Lagos_Boundary*. Save as *Roads_Lagos* and *Waterways_Lagos*.

Coordinate system tip: if ArcGIS Pro asks for a geographic transformation, UTM 31N and WGS84 share the same datum so no transformation is needed — leave it empty.

4.3 Separate drainage lines from roads

Drainage lines include rivers, streams, drains, canals and ditches. Roads are a separate feature class. Use **Select By Attributes** to split them.

- 1 Right-click *Waterways_Lagos* → **Attribute Table**. Confirm the field that carries OSM tags (commonly *waterway* or *fclass* in HDX exports).
- 2 Open **Select By Attributes** and build:

```
fclass IN ('river', 'stream', 'drain', 'canal', 'ditch', 'stream_intermittent')
```

- 1 With the selection active, right-click the layer → **Data > Export Features**. Output = *Drainage_Lines* (in *Lagos_Flood_Risk.gdb*).
- 2 Repeat for roads if you want only major roads. Open *Roads_Lagos* attribute table, **Select By Attributes**:

```
fclass IN ('motorway', 'trunk', 'primary', 'secondary', 'tertiary')
```

- 1 Export the selection as *Roads_Major* (optional, kept for cartography, not used in the risk model).
- 2 Clear selections (*Map > Clear*) before proceeding.

4.4 Buffer drainage channels (50 m and 100 m proximity zones)

We create two buffer rings, dissolve them, then combine them so pixels closer than 50 m are flagged very high proximity risk and pixels 50–100 m are medium proximity risk.

- 1 **Analysis > Tools**, search for **Buffer** (Analysis Tools).

- 2 Run Buffer #1: Input = Drainage_Lines, Distance = **50 Meters**, Side Type = FULL, End Type = ROUND, Dissolve = ALL. Output = Drainage_Buf50.
- 3 Run Buffer #2 identically but Distance = **100 Meters**. Output = Drainage_Buf100.
- 4 Create a ring-only feature class for the 50–100 m band: **Analysis › Erase**. Input = Drainage_Buf100, Erase Features = Drainage_Buf50. Output = Drainage_Ring_50_100.

4.5 Convert proximity zones to a reclassified raster

The risk model is raster-based, so turn the buffers into a 30 m raster scored 1 / 2 / 3.

- 1 Add a short integer field Prox_Rank to both Drainage_Buf50 (set value = **3**) and Drainage_Ring_50_100 (set value = **2**). Use **Calculate Field**.
- 2 **Data Management › General › Merge**: merge the two polygon layers into Drainage_Prox_Poly, keeping Prox_Rank.
- 3 **Conversion Tools › To Raster › Polygon to Raster**: Input = Drainage_Prox_Poly, Value field = Prox_Rank, Cell assignment = CELL_CENTER, Cellsize = 30. Output = Drainage_Prox_Raw.
- 4 Areas outside the buffers are NoData. Reclassify NoData to **1** (low proximity risk) so it can be added in the raster calculator: **Spatial Analyst › Reclass › Reclassify**. Old → New: 1→1, 2→2, 3→3, NoData→1. Output = DrainageProximity_reclassified.

Deliverable of Step 4: a 30 m raster DrainageProximity_reclassified covering the whole Lagos boundary with values 1 (low, >100 m from drainage), 2 (medium, 50–100 m), and 3 (high, <50 m).

Step 5 — Flood Risk Index Calculation

We combine three reclassified rasters with the weights given in the project spec:

```
FloodRisk = (Rainfall_reclassified * 0.4) + (InvElevation_reclassified * 0.4) +  
(DrainageProximity_reclassified * 0.2)
```

5.1 Create InvElevation_reclassified (low ground = higher risk)

The raw DEM scores high for mountains. For flood risk we need the *inverse* — low elevation should be high risk.

- 1 **Spatial Analyst > Reclass > Reclassify** on Lagos_DEM_Fill. Use *Classification > Classify*, Method = **Quantile** or **Natural Breaks (Jenks)**, Classes = 3.
- 2 After Arc proposes three ranges, **invert the new values**. Example (Lagos-typical ranges):

Old value (m)	Meaning	New value
0 – 5	Very low / coastal	3 (High risk)
5 – 20	Low plains	2 (Medium risk)
> 20	Higher ground / uplands	1 (Low risk)

- 1 Save output as InvElevation_reclassified.
- 2 Inspect the result (Symbology → Unique Values) and check that coastal Lagos Island, Lekki lowlands and Makoko show up as class 3.

5.2 Align the three rasters

Before multiplying rasters you want identical extent, cell size, and snap. Because we already set the environments at the top, the three rasters should match. Quickly verify with **Data Management > Raster > Raster Properties** or right-click → Properties > Source. Cell size = 30, rows/cols equal, extent equal.

If any raster is off, run **Resample** (Nearest Neighbor for categorical 1/2/3 codes) to match Lagos_DEM_Fill.

5.3 Run the weighted overlay in Raster Calculator

- 1 **Spatial Analyst Tools > Map Algebra > Raster Calculator**.
- 2 Enter exactly:

```
("Rainfall_reclassified" * 0.4) + ("InvElevation_reclassified" * 0.4) +  
("DrainageProximity_reclassified" * 0.2)
```

- 1 Output = FloodRisk_Index. It will be a continuous raster with values roughly between 1.0 and 3.0.
- 2 Sanity-check the histogram (Symbology pane → Histogram). You should see a roughly normal distribution centred near 2.0.

5.4 Reclassify into Low / Medium / High classes

- 1 **Spatial Analyst > Reclassify** on FloodRisk_Index.
- 2 Use these breakpoints (they match the 40/40/20 weighting):

Input range	Class	New value
1.00 – 1.66	Low risk	1
1.66 – 2.33	Medium risk	2
2.33 – 3.00	High risk	3

- 1 Output = FloodRisk_Class.
- 2 Symbolise with a green → yellow → red ramp (Symbology = Unique Values, 1 = green, 2 = yellow, 3 = red).

5.5 Convert risk raster to polygon

- 1 **Conversion Tools › From Raster › Raster to Polygon.**
- 2 Input = FloodRisk_Class, Field = Value, **Simplify polygons** = checked (reduces staircase edges), **Create multipart features** = unchecked.
- 3 Output = FloodRisk_Polygons. Rename the gridcode field to RiskClass in the attribute table for clarity.
- 4 Add a text field RiskLabel and **Calculate Field** with a Python 3 expression:

```
RiskLabel = ["Low", "Medium", "High"][!RiskClass! - 1]
```

Deliverable of Step 5: FloodRisk_Class (raster) and FloodRisk_Polygons (vector) classifying every 30 m pixel of Lagos as Low, Medium, or High flood risk.

Step 6 — Population Vulnerability Overlay

6.1 Download and prepare WorldPop

- 1 Go to **worldpop.org** › **Open Data** › **Population Counts** › **Unconstrained individual countries 2020 (100 m resolution)** and download `nga_ppp_2020.tif` (Nigeria, WGS84 geographic, 3 arc-seconds $\approx$ 100 m).
- 2 Add the raster to ArcGIS Pro. It is in EPSG:4326.
- 3 **Project Raster** (Data Management › Raster › Projections and Transformations) to WGS 1984 UTM Zone 31N. Resampling = BILINEAR (continuous population surface). Output = `WorldPop_NGA_UTM`.
- 4 **Spatial Analyst** › **Extraction** › **Extract by Mask**. Input = `WorldPop_NGA_UTM`, Mask = `Lagos_Boundary`. Output = `Pop_Lagos`.

Important: WorldPop pixel values are **persons per pixel**, not density. When you resample / project, bilinear conserves the shape of the surface but the cell total changes with the new cell area. Since we only use it as an additive count inside Zonal Statistics (SUM), this is acceptable. If you need strict persons-per-pixel, reproject with Cell size = original and leave resampling as NEAREST.

6.2 Sum population per flood-risk polygon

- 1 **Spatial Analyst** › **Zonal** › **Zonal Statistics as Table**.
- 2 Input feature zone data = `FloodRisk_Polygons`, Zone field = `OBJECTID` (unique per polygon), Input value raster = `Pop_Lagos`, Statistics type = **SUM**. Output table = `Pop_by_RiskPoly`.
- 3 The table now has fields `OBJECTID_1` (matching the polygon `OBJECTID`), `COUNT`, `AREA`, `SUM`. `SUM` = affected population.

6.3 Join the table back to the polygons

- 1 Right-click `FloodRisk_Polygons` → **Joins and Relates** › **Add Join**.
- 2 Input join field = `OBJECTID`, Join table = `Pop_by_RiskPoly`, Output join field = `OBJECTID_1`. Keep all target features.
- 3 Open the attribute table. Each polygon now has a `SUM` column = population living in that flood-risk zone.
- 4 Make the join permanent: right-click layer → **Data** › **Export Features** → `FloodRisk_Pop`.
- 5 Add a numeric field `Pop_Affected` and **Calculate Field** = `!SUM!`. Then remove the join on the original layer (optional cleanup).

6.4 Summarise per risk class and per LGA (optional roll-up)

For the dashboard you usually want one number per LGA, or one number per risk class.

- 1 **Analysis** › **Summary Statistics** on `FloodRisk_Pop`. Statistics field = `Pop_Affected` (SUM), Case field = `RiskClass`. Output = `Pop_per_RiskClass`.
- 2 **Spatial Join** (Analysis › Overlay) `FloodRisk_Pop` onto LGAs (`Lagos_LGA`), Match option = INTERSECT, Merge rule for `Pop_Affected` = SUM. Output = `LGA_Pop_at_Risk`. This is the layer the dashboard charts will point at.

Deliverable of Step 6: `FloodRisk_Pop` (per-polygon affected population) and `LGA_Pop_at_Risk` (per-LGA affected population).

Step 7 — Hotspot Analysis

Optimized Hot Spot Analysis (Getis-Ord Gi*) tells you where statistically significant clusters of high-risk polygons sit, which is exactly what urban planners need for prioritisation.

7.1 Prepare the input field

- 1 Use FloodRisk_Pop as the input. The analysis field should be numeric and continuous; ideally combine risk and exposure. Add a field Risk_Weighted and calculate:

```
!RiskClass! * (!Pop_Affected! + 1)
```

The +1 prevents zero-population polygons from dropping out, and the multiplication means a High-risk polygon with many people dominates the clustering result.

7.2 Run Optimized Hot Spot Analysis

- 1 **Spatial Statistics Tools > Mapping Clusters > Optimized Hot Spot Analysis.**
- 2 Input Features = FloodRisk_Pop. Analysis Field = Risk_Weighted. Output Features = FloodRisk_Hotspots.
- 3 Let the tool pick the scale of analysis automatically — it uses Incremental Spatial Autocorrelation to choose the distance band and reports it in the **Geoprocessing History**. Record this distance in your project notes.

Output fields produced by the tool:

- bullet GiZScore – z-score (positive = hotspot, negative = coldspot).
- bullet GiPValue – significance.
- bullet Gi_Bin – categorical: -3, -2, -1, 0, +1, +2, +3 (99%, 95%, 90% confidence levels).

7.3 Inspect and style the result

- 1 The layer auto-symbolises red for hotspots and blue for coldspots. Leave the default unless you need to match a brand palette.
- 2 Open the attribute table and sort by Gi_Bin descending. Polygons with Gi_Bin = 3 are the priority areas (hotspots at 99% confidence).
- 3 Cross-check against Lagos geography: expected hotspots include Lagos Island, Ajegunle, Ikoro du low-lying wards, Lekki Phase 1 and 2, and parts of Ikeja near Ogun River. If none appear, re-examine the input field.

7.4 Export hotspot layer for the dashboard

- 1 Right-click FloodRisk_Hotspots → **Sharing > Share As Web Layer**. Layer type = **Feature**. Select your ArcGIS Online / Enterprise portal folder and tag it *Lagos, flood, hotspot*. Analyse and Publish.
- 2 Repeat the web-layer publish for FloodRisk_Polygons, LGA_Pop_at_Risk, and the Rainfall_reclassified raster (as a tile layer).
- 3 In ArcGIS Online, open **Dashboards (Instant Apps)**, choose a template (e.g. *Minimalist* or *Dashboard*), and drop in: a map widget, a serial chart of Pop_Affected per LGA, an indicator showing total High-risk population, and a list of hotspot LGAs filtered by Gi_Bin = 3.

Deliverable of Step 7: FloodRisk_Hotspots feature layer published to ArcGIS Online, ready to be dropped into the Instant Apps / Dashboards configuration.

Appendix A — QA Checklist

- bullet All rasters share CRS = WGS 1984 UTM Zone 31N, cell size = 30 m, snapped to Lagos_DEM_Fill.
- bullet No NoData inside the Lagos boundary on any of the three reclassified rasters (would propagate NoData through the raster calculator).
- bullet Flood risk values range 1.0 – 3.0; classified raster has values 1, 2, 3 only.
- bullet Total Pop_Affected across all polygons approximately equals the total of Pop_Lagos zonal sum over the Lagos boundary (sanity check, should be $\approx$ 15 million in 2020).
- bullet Optimized Hot Spot Analysis tool completed without warnings; distance band recorded in the project log.

Appendix B — Common issues

- bullet **Raster Calculator fails with ERROR 010428:** wrap each raster name in quotes exactly as they appear in the Contents pane. Case matters.
- bullet **Buffers look distorted:** data is still in geographic CRS (EPSG:4326). Re-run Project before buffering.
- bullet **Zonal Statistics returns only 1 row:** Zone field is not unique. Use OBJECTID.
- bullet **OSM waterways field name differs:** HDX exports sometimes use waterway; QuickOSM uses fc lass or the raw OSM tag. Inspect the table first.
- bullet **Hot spot tool says ‘too few features’:** you are probably running it on the summarised LGA layer. Use the original FloodRisk_Pop polygons (thousands of features).

Appendix C — Layer inventory produced

Name	Type	Purpose
Drainage_Lines	Feature class (line)	OSM rivers / streams / drains, Lagos-clipped
Drainage_Buf50 / Buf100	Feature class (polygon)	Proximity buffers
DrainageProximity_reclassified	Raster (30 m)	Step 4 output; values 1 / 2 / 3
InvElevation_reclassified	Raster (30 m)	Step 5 input; low ground = 3
FloodRisk_Index	Raster (30 m, float)	Weighted sum output
FloodRisk_Class	Raster (30 m, int)	Reclassified 1 / 2 / 3
FloodRisk_Polygons	Feature class (polygon)	Vector version for joins
Pop_Lagos	Raster (100 m)	WorldPop clipped to Lagos
FloodRisk_Pop	Feature class (polygon)	Per-polygon affected population
LGA_Pop_at_Risk	Feature class (polygon)	Per-LGA roll-up for the dashboard
FloodRisk_Hotspots	Feature class (polygon)	Optimized Hot Spot Analysis result

Generated with Kiro for the Lagos Flood Risk Monitoring Dashboard project. Verify layer names against your own geodatabase before running each tool.